Reversible change detectives

Teacher guide and worked answers

Describe melting, freezing and dissolving and explain how the starting substances can be recovered in these examples.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C33

Describe reversible changes (dissolving, melting, freezing).

Intentional revisit and extension of earlier solubility work; separating dissolving from melting adds depth. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Identify solid and liquid states.

Know that dissolved substances may no longer be visible.

Success evidence

Distinguish a change of state from mixing with a solvent.

Explain that dissolved sugar remains present.

Choose a suitable reverse process and state what is recovered.

Equipment

Constructed evidence cards, selected route, pencils; optional particle tokens.

Preparation

The default lesson is paper-based. Keep melting and dissolving verbs separate.

Explain that particle circles are symbols, not visible grains or a to-scale molecular diagram.

Practical care

No heating, tasting, sealed-container warming or freezing practical is needed.

Any later practical requires separate preparation; do not heat sugar to recover it in this lesson.

Ready to teach alternative

Use supplied observations and particle tokens to compare processes without heat, food handling or a time-dependent trial.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The crystal has disappeared from view

Slide	Stage	Minutes	Focus
2	Retrieval	2-5	State or mixture?
3	I do	5-13	I identify a change of state
4	I do	Within stage	I count what the model retains
5	I do	Within stage	I choose a route back
6	We do	13-21	Process evidence tribunal
7	We do	Within stage	Choose a reverse process
8	We do	Within stage	Write a precise verdict
9	Hinge	21-24	Sugar is stirred into room-temperature water
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The crystal has disappeared from view

Present the question with paper evidence. No tasting is allowed or needed to identify a dissolved substance.

2 Retrieval State or mixture?

Accurate examples; no; grams or kilograms. Do not assume all clear liquids are pure water.

3 I do I identify a change of state

Think aloud: I track water before and after. A second substance is not required for the melting step. Energy transfer matters; warming does not need to continue raising temperature during the phase change. Numerical phase temperatures are not needed.

4 I do I count what the model retains

Shared I do time. Model the four orange symbols before and after: same number, new distribution. Blue and orange symbol sizes are schematic. Sucrose molecules are not broken into atoms in this dissolving example.

5 I do I choose a route back

Shared I do time. Explain that the recovery observation spans days and is constructed. No rapid or heated sugar recovery is instructed. This lesson distinguishes concepts; next lesson chooses specific recovery methods.

6 We do Process evidence tribunal

A contains melting and freezing; B dissolving. C supports no loss of mass from the sealed system; D shows sugar can be recovered. Mass alone does not identify every substance.

7 We do Choose a reverse process

Shared We do time. Cooling freezes water; water evaporation can recover sugar from solution. Simply putting sugar solution in a freezer is not the same as cleanly recovering both components.

8 We do Write a precise verdict

Sugar molecules remain in a mixture with water. The result is solution, not conversion of sugar into water. Keep molecular detail age-appropriate.

Reveal: Sugar remains mixed through the water. Removing water can leave sugar behind.

9 Hinge Sugar is stirred into room-temperature water

A is correct. B incorrectly calls dissolving a change of state; C incorrectly claims that the sugar vanishes.

A. The sugar dissolved and remains in the solution. - The particles are distributed through the solvent.

B. The sugar melted into pure liquid sugar. - This is mixing with water to form a solution, not melting.

C. The sugar stopped existing. - The mass and recovery evidence contradict disappearance of the substance.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

Melting changes the state of a substance; dissolving makes a mixture. The sugar remains distributed through the water. Evaporating the water over time can recover sugar. E4 asks for the reverse of melting: freezing.

Answers and assessment

R1-R3

Valid solid/liquid examples; invisible does not mean absent; grams or kilograms.

W1-W3 / reverse routes

A melting and freezing; B dissolving. C shows unchanged total mass in a closed system; D shows recoverable sugar after water leaves. Enough cooling refreezes melted ice; evaporation removes water to recover sugar. W3 sugar remains distributed in a mixture with water, not converted into water.

Hinge A-C

A correct. B confuses dissolving with melting. C incorrectly claims the substance ceases to exist.

S1-S6

Melting; freezing; dissolving; sugar remains mixed through the water; evaporation removes water leaving sugar; no, simple open evaporation does not collect water.

N1-N4

Melting: ice → liquid water; freezing: liquid water → ice; dissolving: sugar plus water → sugar solution. N2 state change versus mixture formation with solvent. N3 same total mass plus sugar recovery contradict sugar vanishing; neither observation requires tasting. N4 allow water to evaporate over time; sugar remains, water is not captured by open evaporation.

T1-T4

T1 water remains water while state/arrangement changes; sugar and water remain present as a mixture, sugar molecules distributed among water molecules. T2 constant mass indicates no measured mass loss from the system but does not identify all substances; recovery specifically supports sugar remaining. T3 cooling refreezes water but does not universally separate every mixture into original components. T4 open evaporation leaves solid while water vapour escapes; recovering water requires capturing vapour and condensing it, explored next lesson.

E1-E4 / assessment

State change versus mixing; sugar remains distributed; evaporate water; freezing. Secure: distinguishes processes and gives an appropriate recovery route with its target. Re-teach by drawing one water substance versus two components in a solution. Reflection accepts a relevant alternative target.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This is an intentional deeper revisit of earlier solubility work: the new focus separates dissolving from state change and distinguishes recovery of one component from both.

All particle pictures are selective models, with unchanged symbol counts but no claim of exact molecular size or concentration. Sugar dissolving is chosen so whole sucrose molecules remain intact.

Paper evidence reproduces every interactive decision. Evaporation over days is a constructed record, not a process claimed to finish during the 40-minute lesson.

Access and responsive teaching

Supported

Classify three cases and use a particle word bank.

Standard

Explain process differences and use mass/recovery evidence.

Stretch

Evaluate a disappearing-sugar claim and distinguish recovery of one component from both.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

Dissolving is the same as melting.

Melting changes one substance from solid to liquid; dissolving forms a mixture with a solvent.

Check: Ask whether a second substance is involved.

Clear sugar solution contains no sugar.

Sugar molecules remain mixed among water molecules even when crystals are not visible.

Check: Ask what the recovery record shows.

Cooling a solution always gives back all its original separate substances.

The useful reverse process depends on the mixture and target; removing water can recover dissolved solid in this example.

Check: Ask which substance must be removed or collected.

Word	Meaning
melting	A substance changing from solid to liquid.
freezing	A substance changing from liquid to solid.
dissolving	A substance mixing through a solvent to form a solution.
solution	A mixture in which the dissolved substance is distributed through a solvent.
reversible	Able to be changed back in the stated example or conditions.
evaporation	Liquid changing to gas at its surface.

Scientific references

American Chemical Society: Why does water dissolve sugar?

Checked September 2026. Sugar molecules separate from one another and mix among water molecules without the sucrose molecules being destroyed.

American Chemical Society: Changes of state

Checked September 2026. Melting, freezing, evaporation and condensation; practicals here are replaced by original paper evidence.

Royal Society of Chemistry: Separation techniques

Checked September 2026. Filtration, evaporation and separation misconceptions. Classroom activities are original and include a complete paper route.